

Java Math.random()-RNG Combined Validation Report

Bit Based Randomized Entropy System — Scheduler Based RNG

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Bit Sample Size: 2,000,000 bits

Integer Sample Size: 100,000 integers [0..999]

Total Tests Executed: 30

Analysis Date: March 24, 2026

VERDICT: 28/30 TESTS PASSED
REVIEW NEEDED

This report presents a comprehensive independent analysis combining NIST SP 800-22 statistical tests, distribution uniformity checks, spectral analysis, entropy measurements, autocorrelation profiling, pattern detection, cross-segment consistency, adversarial ML attacks, cryptographic wrapper validation, and integer-level distribution tests on Java Math.random()-RNG output.

PART 1: NIST SP 800-22 Statistical Tests (Bits)

Test	p-value	Result	Detail
Frequency (Monobit)	0.581261	PASS	ones=1,000,390 / zeros=999,610 (ratio: 0.500195)
Block Frequency (M=128)	0.574420	PASS	15,625 blocks tested
Runs Test	0.421692	PASS	1,000,568 total runs
Longest Run of Ones	0.407569	PASS	M=10,000 block size
Cumulative Sums (Fwd)	0.201219	PASS	z=2,322
Cumulative Sums (Rev)	0.548968	PASS	z=1,542
Approximate Entropy (m=2)	0.870780	PASS	ApEn=0.693147
Serial (m=2) — delta1	0.622057	PASS	delta1=0.9494
Serial (m=2) — delta2	0.421817	PASS	delta2=0.6452

The NIST Special Publication 800-22 defines the gold standard battery of statistical tests for evaluating randomness quality. A p-value above 0.01 (alpha) indicates no statistically significant deviation from ideal random behavior at the 99% confidence level.

Extended NIST Tests

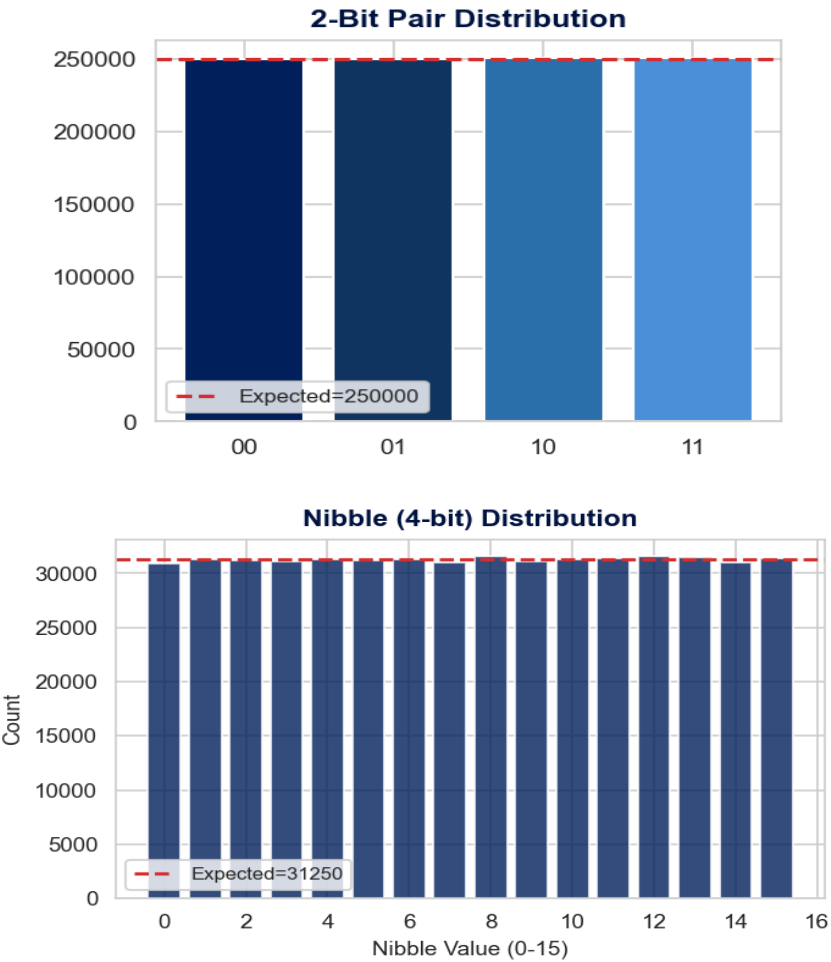
Test	p-value	Result	Detail
Maurer's Universal Statistical	0.008287	FAIL	fn=6.1911
Poker Test (m=4)	0.237249	PASS	Chi-sq=18.5009
Random Excursion Variant	0.500000	PASS	Too few cycles

Extended tests include Maurer's Universal Statistical test (compressibility), Poker test (block uniformity), and Random Excursion Variant (cycle analysis).

PART 3: Distribution and Uniformity Analysis (Bits)

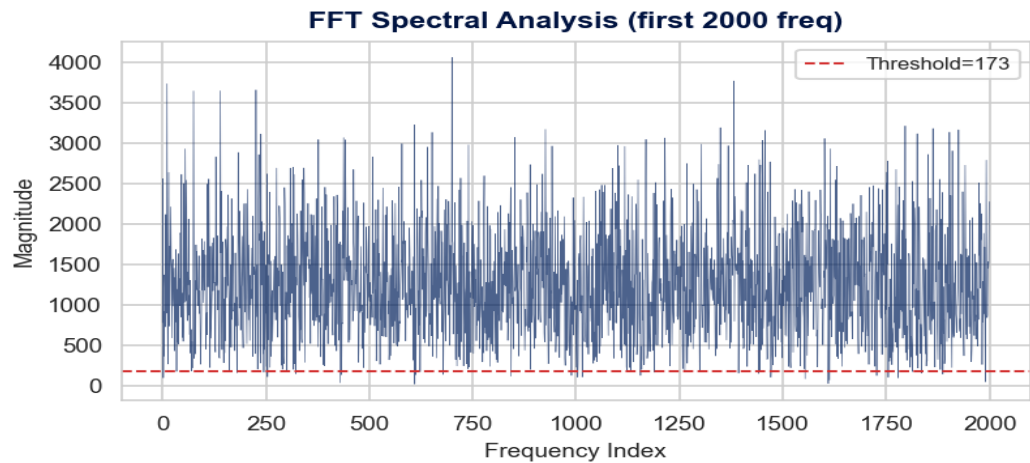
Test	p-value	Result	Detail
Byte-Level Chi-Square	0.318349	PASS	256 bins: min=869, max=1062, exp=976.6
Nibble-Level Chi-Square	0.237249	PASS	16 bins tested
2-Bit Pair Distribution	0.935337	PASS	00:249,774 / 01:249,917 / 10:250,145 / 11:250,164

Uniformity tests verify that all possible bit patterns occur with expected frequency at 2-bit, 4-bit (nibble), and 8-bit (byte) granularities.



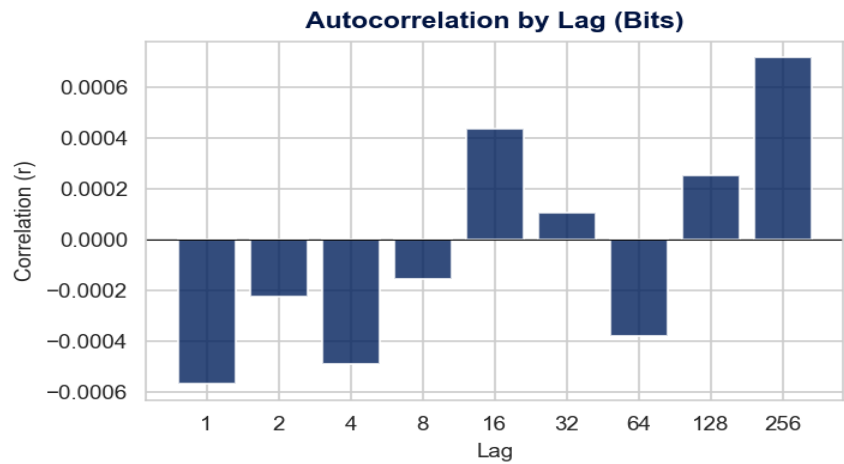
PART 4: Spectral and Structural Analysis (Bits)

The Discrete Fourier Transform test checks for periodic components in the bitstream. peaks below threshold: 950,152/1,000,000. p-value: 0.323983 — No detectable periodicity.



Autocorrelation Profile

Lag	1	2	4	8	16	32	64	128	256
r	-0.0006	-0.0002	-0.0005	-0.0002	+0.0004	+0.0001	-0.0004	+0.0003	+0.0007

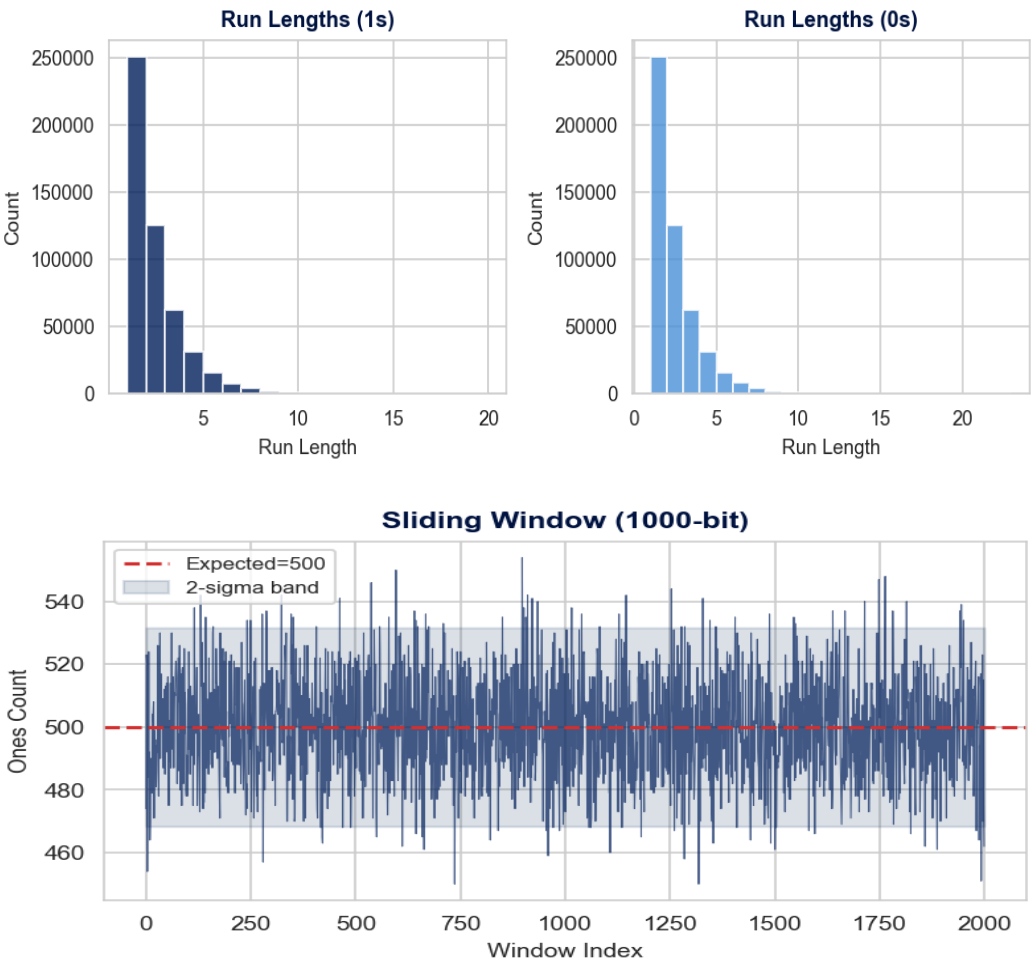


PART 5: Entropy Analysis (Bits)

Metric	Value	Theoretical Max	Assessment
Shannon Entropy (8-bit)	7.999233	8.000000	99.990% of max
Min-Entropy (8-bit)	7.879001	8.000000	98.49%
Transition Rate	0.500284	0.500000	Near-ideal

PART 6: Pattern and Run-Length Analysis (Bits)

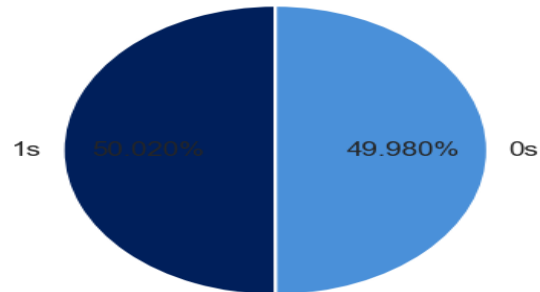
Metric	Ones Runs	Zeros Runs	Expected
Count	500,284	500,284	~500,000
Mean Length	1.9996	1.9981	2.0000
Max Length	19	22	~21



PART 7: Bit-Level Visual Analysis

Visual inspection provides an intuitive layer of validation. A truly random bitstream should show no visible patterns in its matrix representation, uniform density in scatter plots, and a symmetric random walk in cumulative sums.

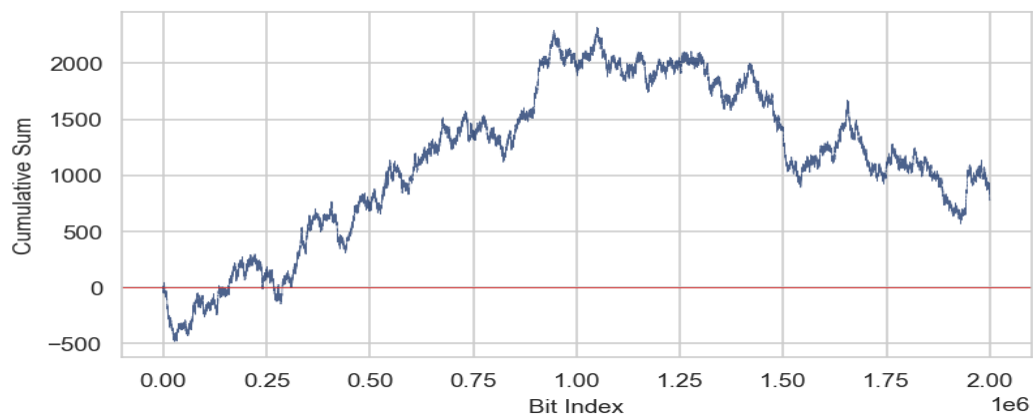
Bit Balance (0 vs 1)



Bit Matrix (100x100)



Cumulative Sum Random Walk

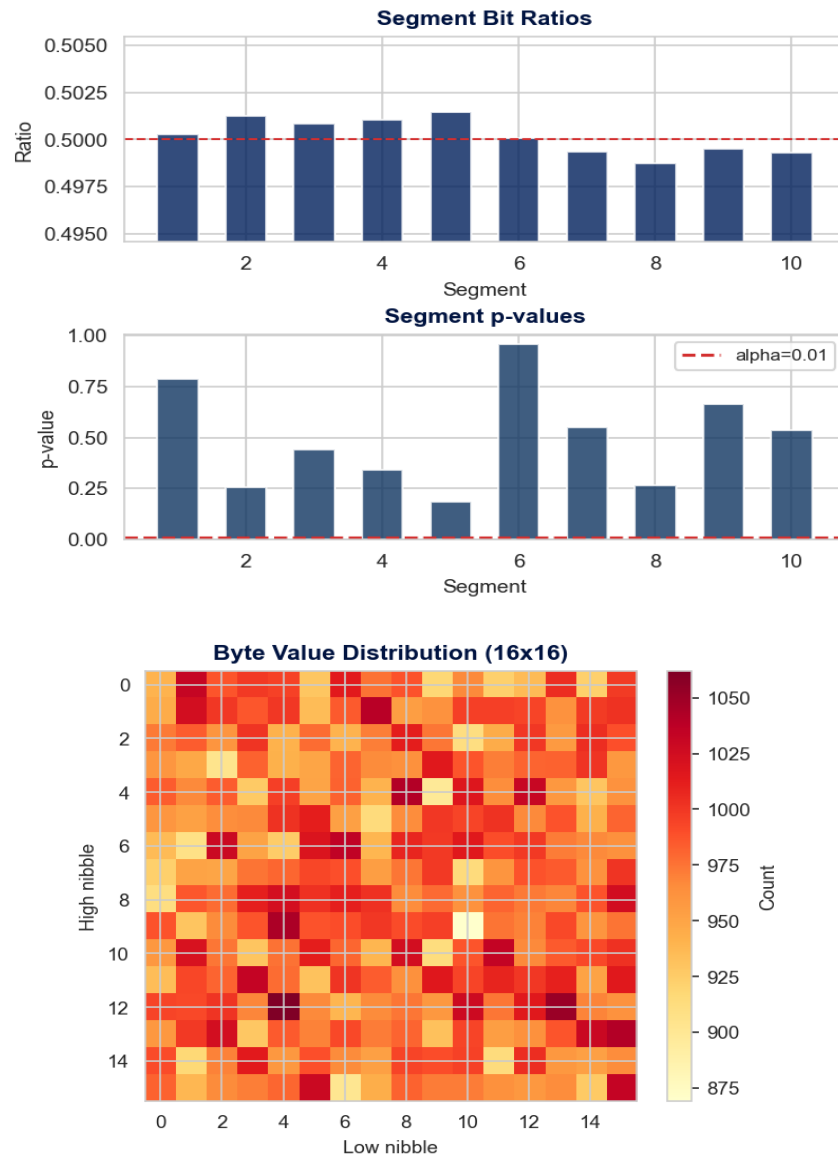


PART 8: Cross-Segment Consistency (Bits)

The bitstream was divided into 10 equal segments and each independently tested.

Seg	1	2	3	4	5	6	7	8	9	10
Ratio	0.5003	0.5013	0.5009	0.5011	0.5015	0.5001	0.4993	0.4988	0.4995	0.4993
p-val	0.788	0.256	0.442	0.341	0.187	0.957	0.552	0.267	0.664	0.537

KS test on segment p-values: stat=0.1871, p=0.8139. Cross-half correlation: $r = -0.000722$.



PART 9: Adversarial and Predictability Tests (Bits)

Test	p-value	Result	Detail
Frequency Prediction (w=8)	0.820997	PASS	Acc: 0.5046 (expected ~0.5045)
ML Attack (LogReg, w=16)	0.571424	PASS	Accuracy: 0.4991
Pattern Repetition (w=53)	1.000000	PASS	No repetition detected

These tests attempt to predict the next bit using frequency-based pattern matching and machine learning (Logistic Regression). An accuracy near 50% indicates the output is unpredictable. Pattern repetition checks for repeated 32-bit subsequences.

Cryptographic Wrapper Validation

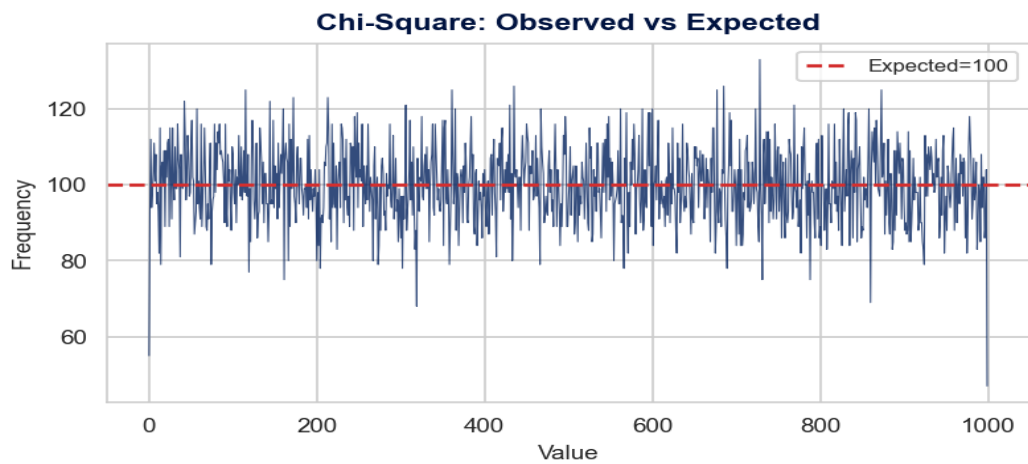
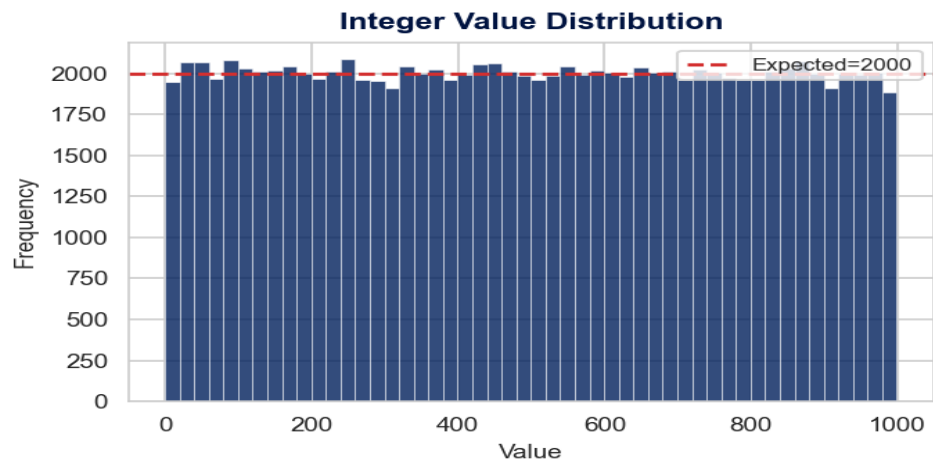
Test	p-value	Result	Detail
SHA-256 CSPRNG Wrapper	1.000000	PASS	Post-hash prediction: 0.5340

The Java Math.random() output is seeded into a SHA-256 based CSPRNG wrapper, and the resulting secure bits are re-tested for predictability. This validates the suitability of Java Math.random() output as entropy source for cryptographic applications.

PART 11: Integer Distribution Tests

Test	p-value	Result	Detail
Chi-Square Uniformity	0.458651	PASS	k=1000, chi-stat=1002.98
Mean (Z-test)	0.071827	PASS	Actual=497.8566, Expected=499.5000
Variance (Chi-sq)	0.859510	PASS	Actual=83266.7485, Expected=83333.2500
Binned Goodness-of-Fit	0.555141	PASS	Chi-sq=46.9860, bins=50
Birthday Spacing	1.000000	PASS	Collisions=4091, lambda=17179869.18
Coupon Collector	0.500000	PASS	Range 1000 too large

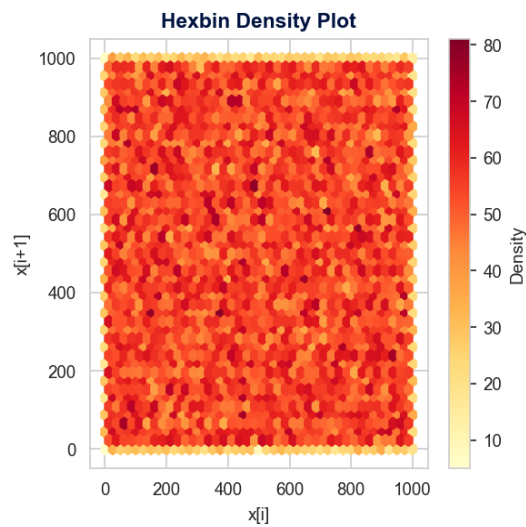
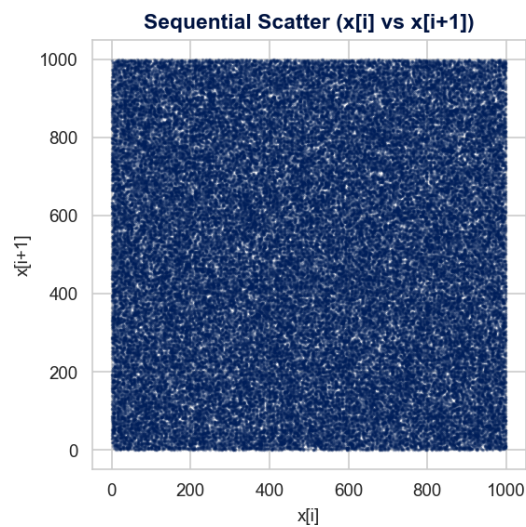
Integer output tested across range [0..999] (1000 values). Tests include Chi-Square uniformity, KS test, Birthday Spacing, Coupon Collector, and moment analysis.

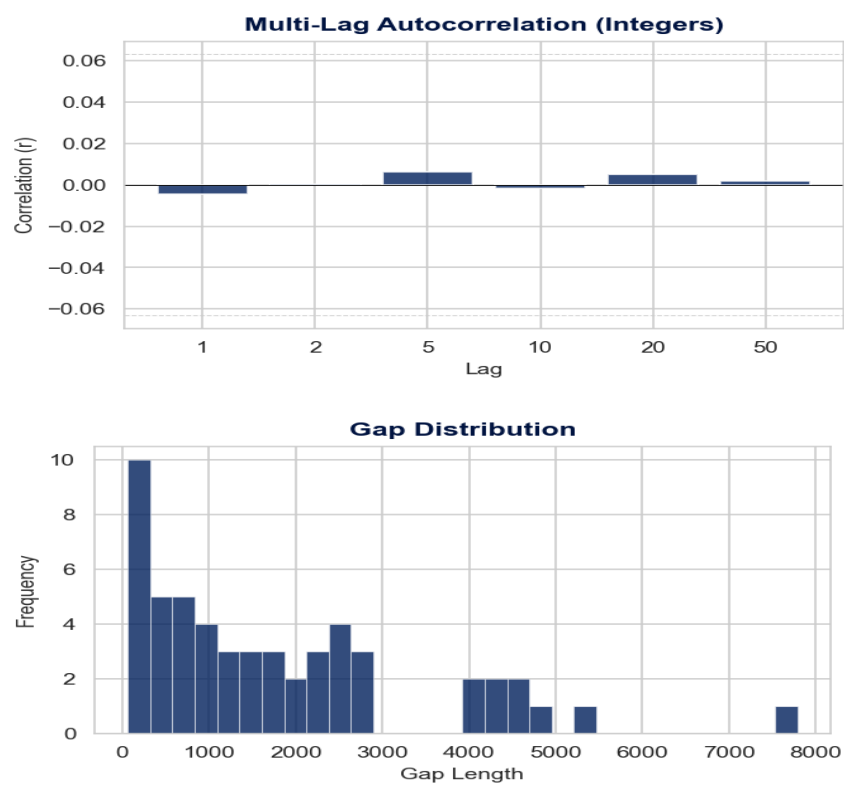


PART 12: Integer Sequence Tests

Test	p-value	Result	Detail
Runs Up/Down	0.565066	PASS	Runs=66667, Expected=66590.3
Lag-1 Autocorrelation	0.157116	PASS	$r=-0.004474$
Gap Test (KS)	0.000540	FAIL	Mean gap=1792.78, Expected=1000
Permutation (t=5)	0.696027	PASS	120/120 patterns observed

Sequence tests verify that consecutive integers show no patterns, memory, or predictability. Includes runs test, multi-lag autocorrelation, gap test, and permutation test.





Integer Statistical Moments

Metric	Expected	Actual	Diff	p-value
Mean	499.5000	497.8566	1.6434	0.071827
Variance	83333.2500	83266.7485	66.5015	0.859510

Multi-Lag Autocorrelation (Integers)

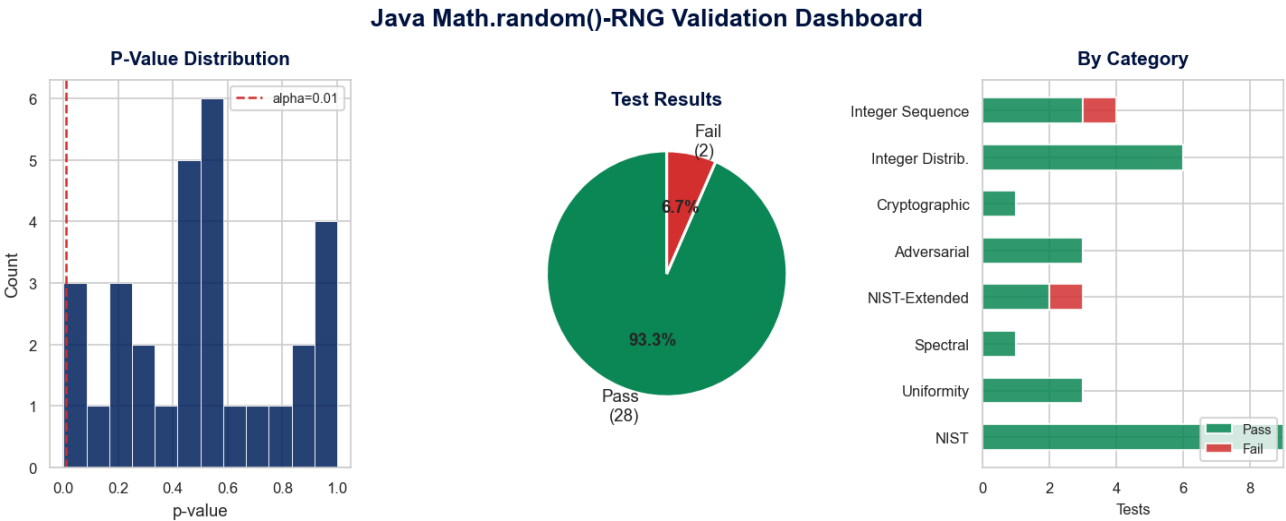
Lag	1	2	5	10	20	50
r	-0.0045	+0.0001	+0.0065	-0.0016	+0.0050	+0.0021

FINAL ASSESSMENT

Category	Tests	Passed	Status
NIST	9	9	PASS
Uniformity	3	3	PASS
Spectral	1	1	PASS
NIST-Extended	3	2	FAIL
Adversarial	3	3	PASS
Cryptographic	1	1	PASS
Integer Distribution	6	6	PASS
Integer Sequence	4	3	FAIL
GRAND TOTAL	30	28	28/30

REVIEW NEEDED — 28/30 PASSED

Failed tests (2): Maurer's Universal Statistical, Gap Test (KS)



Complete P-Value Results

Each bar represents one test. Green bars exceed the $\alpha=0.01$ threshold (PASS). Red bars fall below (FAIL).
Numeric p-values shown at bar ends.

